AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS 2022

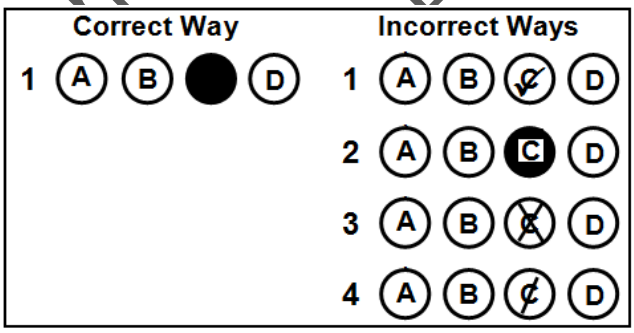
Business Statistics

Time: 1 hour 20 minutes Marks: 40

Annual Examinations 2022 for Learning & Teaching Only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 40 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. The marks obtained on the 40 MCQs will be equated to the total marks of 50 for the theory examination results.
8. You may use a scientific calculator if you wish.

1. Which of the given statements is FALSE for descriptive statistics?

   A. It summarises the sample data.
   B. It presents data in meaningful way.
   C. It draws conclusion about population.
   D. It represents data through graphs and tables.

2. An example of discrete variable is a person's

   A. weight.
   B. height.
   C. age in years.
   D. body temperature.

3. Which of the following statements is TRUE for sample and population**?**

   A. Population is a subset of sample.
   B. Population is a complete set.
   C. A static describes the whole population.
   D. A characteristic that describes a sample is called parameter.

4. The given table shows the distribution of the ages of participants in a hockey tournament.

Ages of Participants in a Hockey Tournament	**Number of Participants**
10 – 13	6
14 – 17	8
18 – 21	11
22 – 25	3
26 – 29	2

The percentage of the participants aged between 26 to 29 years is

A. 2.00%.
B. 6.67%.
C. 93.33%.
D. 98.00%.

Annual Examinations 2022 for Teaching & Learning Only

Use the given table to answer Q.5 and Q.6.

The table gives information about the number of employees in a certain organisation and distance between their homes and office.

Distance (km)	Number of Employees
1 – 4	15
5 – 8	22
9 – 12	45
13 – 16	23
17 – 20	10

5. The class interval for the given data is

   A. 3
   B. 4
   C. 5
   D. 19

6. The class boundary associated to the highest frequency for the given data is

   A. 8.5 – 11.5
   B. 9.5 – 12.5
   C. 8.5 – 12.5
   D. 9.5 – 11.5

7. The given frequency distribution represents the number of pets owned by the students of a class.

Number of Pets	Number of Students	Cumulative Frequency
0	8	8
1 – 2	7	?
3 – 4	9	24
5 – 6	5	29
7 – 8	1	?

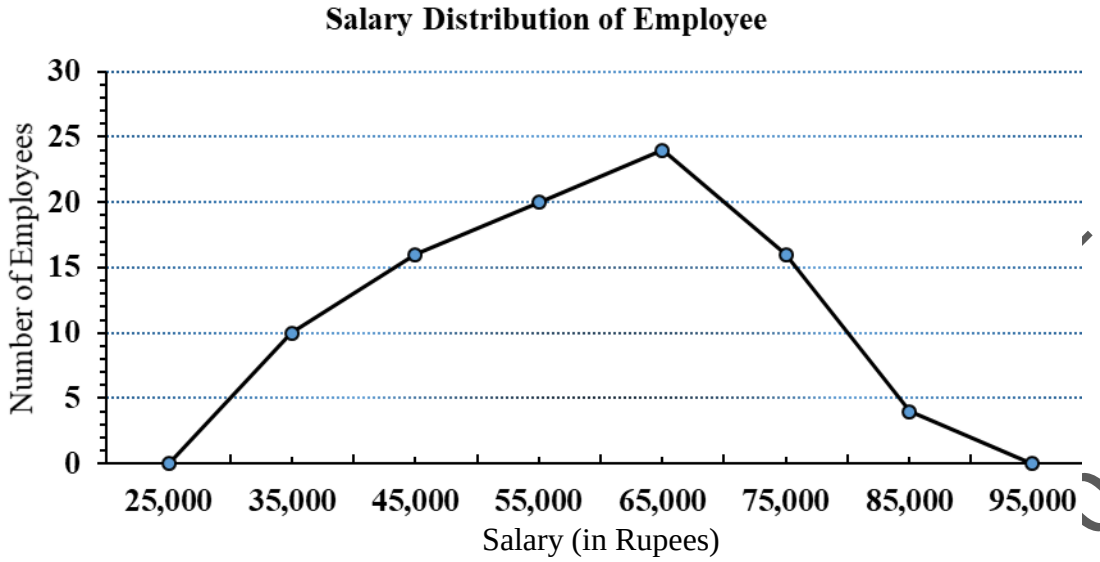
The number of students who owned more than 4 pets is

   A. 5
   B. 6
   C. 9
   D. 14

Annual Examinations 2022
for
Teaching & Learning Only

Use the given frequency polygon to answer Q.8 and Q.9.

The given frequency polygon represents the salary distribution of the employees in an organisation.



8. The total number of employees in the organisation are

   A. 24
   B. 60
   C. 86
   D. 90

9. The minimum salary in the organisation is

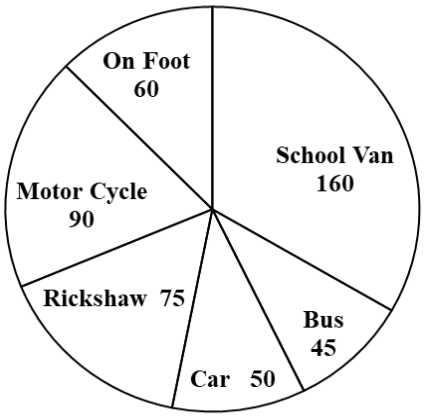
   A. Rs 25,000
   B. Rs 30,000
   C. Rs 35,000
   D. Rs 40,000

Annual Examination for Teaching & Learning Only

Use the given pie chart to answer Q.10, Q.11 and Q.12.

For a school, the given pie chart represents the number of students and their modes of travelling to their school.

Modes of Travelling



10. The percentage of students who travelled to the school by car or motorcycle is

    A. 29.17%.
    B. 26.04%.
    C. 31.25%.
    D. 33.75%.

11. The central angle of the sector representing the number of students travelling to school on foot is

    A. 12.5^{o}
    B. 16.6^{o}
    C. 45°
    D. 80°

12. The central angle of the sector, representing the number of students who DO NOT travel by school van is

    A. 33°
    B. 120°
    C. 150°
    D. 240°

13. The arithmetic mean of 5, 6, 6, 6, 5, 6, 5, 5 is

    A. 5
    B. 5.5
    C. 6
    D. 6.5

Annual Examinations 2022 for Teaching & Learning Only

14. The arithmetic mean of a, a, a, b, b is 10. If $b = 5a$, then the value of a will be

A. $\frac{50}{13}$
B. $\frac{10}{13}$
C. $\frac{25}{13}$
D. 10

15. If the median of 5, 7, 8, c, 10, 10, 12 and 13 is 9.5, then the value of c will be

A. 8.5
B. 9
C. 9.5
D. 10

16. If the arithmetic mean of a, b, c, c and b is 13, then the arithmetic mean of $2a-13$, $2b-13$, $2c-13$, $2c-13$ and $2b-13$

A. is 0
B. is 13
C. is 26
D. cannot be determined

17. For the data 1, 2, 3, 3, 4, 4, 6 and 7, the second quartile (Q_2) is

A. 3
B. 3.5
C. 4
D. 4.5

18. For the data 1, 2, 3, 3, 4, 4, 6 and 7, the third quartile (Q_3) is

A. 4
B. 4.5
C. 5
D. 6

Annual Examinations 2022 for Teaching & Learning Only

Use the given table to answer Q.19, Q.20 and Q.21.

A badminton club gathered information about the weights of its members and recorded the information in the given table.

Weight (kg)	Number of Members	Cumulative Frequency
40 – 49	7	7
50 – 59	18	25
60 – 69	36	61
70 – 79	29	90
80 – 89	10	100

19. The lower class boundary of the median class is

A. 59
B. 59.5
C. 60
D. 60.5

20. The lower class boundary of the third quartile (Q_3) class is

A. 69
B. 69.5
C. 70
D. 70.5

21. How many members of the club weigh less than 70 kg?

A. 36
B. 39
C. 61
D. 69

Annual Examinations 2022 for Teaching & Learning Only

22. The organisers of a swimming competition recorded the data of the participants in the given table.

Ages of Participants in a Swimming Competition (in years)	Number of Participants
10 – 16	16
17 – 23	18
24 – 30	56
31 – 37	13
38 – 44	7

From the given table, the cumulative frequency preceding the median age group will be

A. 18
B. 34
C. 56
D. 90

23. For the data 15, 16, 17, 17, 18, 18 and 19, the first quartile (Q_1) is

A. 16
B. 16.5
C. 17
D. 17.5

24. The variance in the data $1, -1, 1, -1, -1$ and 1 is

A. -1
B. 0
C. 1
D. 2

25. The variance in the data $1-a, -1-a, 1-a, -1-a, -1-a$ and $1-a$ is

A. 0
B. $1-a$
C. 1
D. $(1-a)^2$

26. The standard deviation of the data $a, -a, a, -a, -a$ and a is

A. 0
B. $\sqrt{a}$
C. a
D. a^2

Annual Examinations 2022 for Teaching & Learning Only

27. The standard deviation of the data $2x+a, 2x+a, 2x+a, 2x+a, 2x+a$ and $2x+a$ is

A. 0
B. $2x$
C. a
D. $2x+a$

28. The given table represents the information about number of hours the employees of a company have worked in a certain week.

Working Hours	**Number of Employees**	***x***	***fx***	***fx²***
25 – 27	5	26	130	3,380
28 – 30	10	29	290	8,410
31 – 33	23	32	736	23,552
34 – 36	12	35	420	*a*
37 – 39	7	38	266	10,108
40 – 42	3	41	*b*	5,043
Total	**60**	**-**	**1,965**	**65,193**

The variance in the working hours of the employees is

A. 3.7399
B. 13.9875
C. 32.4499
D. 1053.8

29. If in a data set $\sum x^2 = 10$, $\sum x = 6$ and $n = 5$, then its standard deviation will be

A. 0.56
B. 0.75
C. 0.80
D. 0.89

Annual Examinations 2022.
For Teaching & Learning only

Use the given information to answer Q.30 and Q.31.

For two sets of data the information is presented in the following table.

	Sum of Data $\sum X$	**Mean of Data** $\overline{X}$	**Standard Deviation of Data** *S.D*
Data Set I	20	4	1.095
Data Set II	25	5	0.632

30. The coefficient of variation of the **Data Set II** is

A. 12.64%.
B. 21.90%.
C. 27.37%.
D. 791.14%.

31. The variance in the **Data Set II** is

A. 1.046
B. 1.199
C. 0.399
D. 0.795

32. If the repetition of digits is NOT allowed, then how many 3 digit odd numbers can be formed using digits 1, 2, 3, 4 and 5?

A. 36
B. 28
C. 60
D. 75

33. If two dice are rolled together, then the possible number of ways to get a sum of 8 on the dice will be

(**Note**: Plural of die is dice.)

A. 3
B. 4
C. 5
D. 6

Annual Examinations 2022 for Teaching & Learning Only

34. There are 5 boys and 3 girls in a group of children. The number of ways in which 3 children can be selected from the group is

A. 3
B. 15
C. 56
D. 336

35. There are 5 boys and 3 girls in a group of children. Three children are to be selected from the group. In how many different ways they can be selected if one particular girl should be in the group?

A. 7
B. 10
C. 21
D. 42

36. If a die is rolled and a coin is tossed, then the probability of getting a head on the coin and even number on the die will be

A. $\frac{1}{2}$.
B. $\frac{1}{3}$.
C. $\frac{1}{4}$.
D. $\frac{2}{3}$.

37. How many different arrangements of the letters of the word "ATHLETIC" are possible, if vowels always come together?

A. $\frac{3!\times 5!}{2!}$
B. $\frac{3!\times 6!}{2!}$
C. $3!\times 5!$
D. $3!\times 6!$

Annual Examinations 2022 for Teaching & Learning Only

38. If a die is rolled and two coins are tossed simultaneously, then the probability of getting heads on both the coins and an even number on the die will be

A. $\frac{1}{24}$.

B. $\frac{3}{24}$.

C. $\frac{4}{24}$.

D. $\frac{6}{24}$.

39. There are 10 boys and 18 girls in a class. If two students are selected at random, then the probability that both are boys will be

A. $\frac{10}{28} \times \frac{10}{28}$.

B. $\frac{10}{28} \times \frac{10}{27}$.

C. $\frac{10}{28} \times \frac{9}{28}$.

D. $\frac{10}{28} \times \frac{9}{27}$.

40. A and B are two independent and mutually non-exclusive events. If $P(A) = x$ and $P(B) = x$, then $P(A \cap B)$ will be

A. 0

B. x

C. x^2

D. $2x$

Annual Examinations 2022 for Teaching & Learning Only

END OF PAPER

Please use this page for rough work

Annual Examinations 2022
for
Teaching & Learning Only

Please use this page for rough work

Annual Examinations 2022
for
Teaching & Learning Only

Please use this page for rough work

Annual Examinations 2022
for
Teaching & Learning Only

Please use this page for rough work

Annual Examinations 2022
for
Teaching & Learning Only